

AMENDMENT NO. 1 AUGUST 1987

TO

**IS:2416-1986 SPECIFICATION FOR BOILER AND
SUPERHEATER TUBES FOR MARINE AND
NAVAL PURPOSES**

(First Revision)

(Page 6, clause 5.1) - Substitute the following for the existing clause:

"5.1 The tubes shall be manufactured from steel produced by open hearth, an electric arc or any of the oxygen processes. The chemical composition of each heat when analyzed shall conform to the following analysis:

Carbon	-	0.08 to 0.25 percent
Manganese	-	0.35 to 1.40 percent
Silicon	-	0.35 percent, <u>Max</u>
Phosphorus	-	0.050 percent, <u>Max</u>
Sulphur	-	0.050 percent, <u>Max</u>

Steel produced by the oxygen process shall, in addition, contain not more than 0.009 percent nitrogen.

5.1.1 For alloy steels the tramp elements shall be restricted to the following upper limits:

Copper	-	0.25 percent
Tin	-	0.03 percent
Nickel	-	0.3 percent

Soluble aluminium	-	0.02 percent
Chromium	-	2.25 percent, <u>Max</u>
Molybdenum	-	1.0 percent, <u>Max</u>

5.1.2 The permissible deviations from the specified chemical composition shall be as given in Table 1.

5.1.3 The analysis of steel shall be carried out either by the method specified in IS:228 'Methods for chemical analysis of steels' and its relevant parts or any other established instrumental/chemical method. In case of dispute the procedure given in IS:228 and its relevant parts shall be referee method. However, where the method is not given in IS:228 and its relevant parts, the referee method shall be agreed to between the purchaser and the manufacturer."

(SMDC 22)

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